The Electronic Coupling Between Anthracene Isomers and CdSe Nanoplatelets: Photon Upconversion

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Introduction

Photon Upconversion:

Converting two or more low energy photons into a high energy photon, involving:

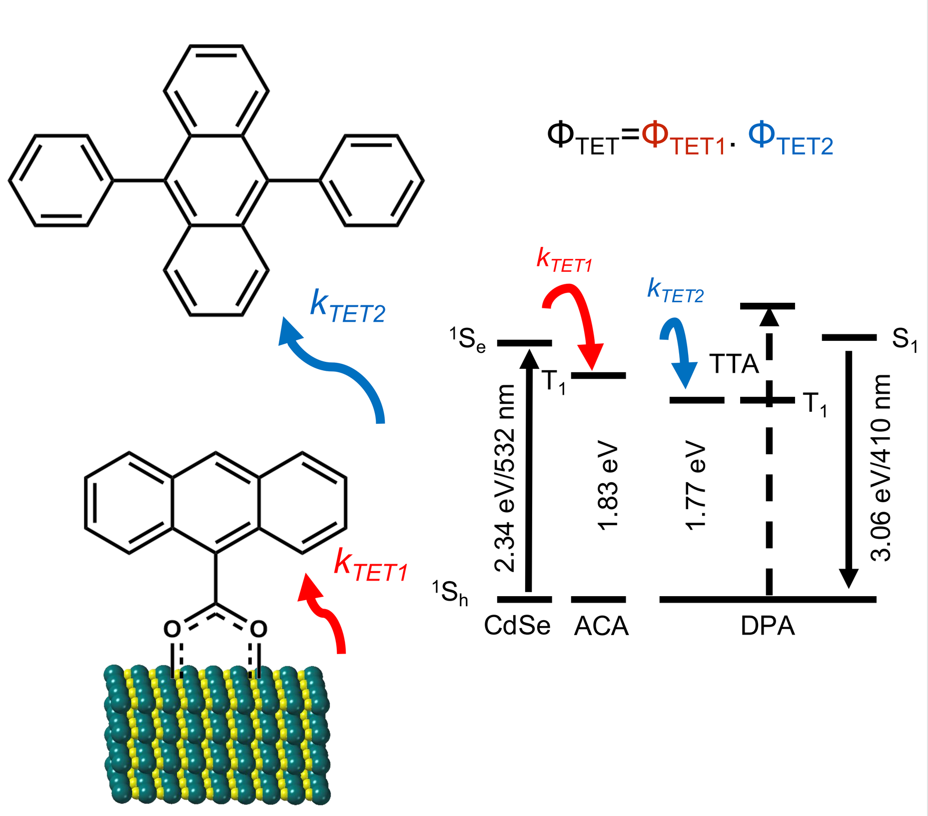
- TET1: the first triplet energy transfer (TET) step from nanoplatelets to the surface bound mediator
- TET 2: the second TET step from the mediator to the emitter in the solution
- Applications: energy conversion, photocatalysis, and bioimaging

This work:
A Hybrid Platform for Photon Upconversion

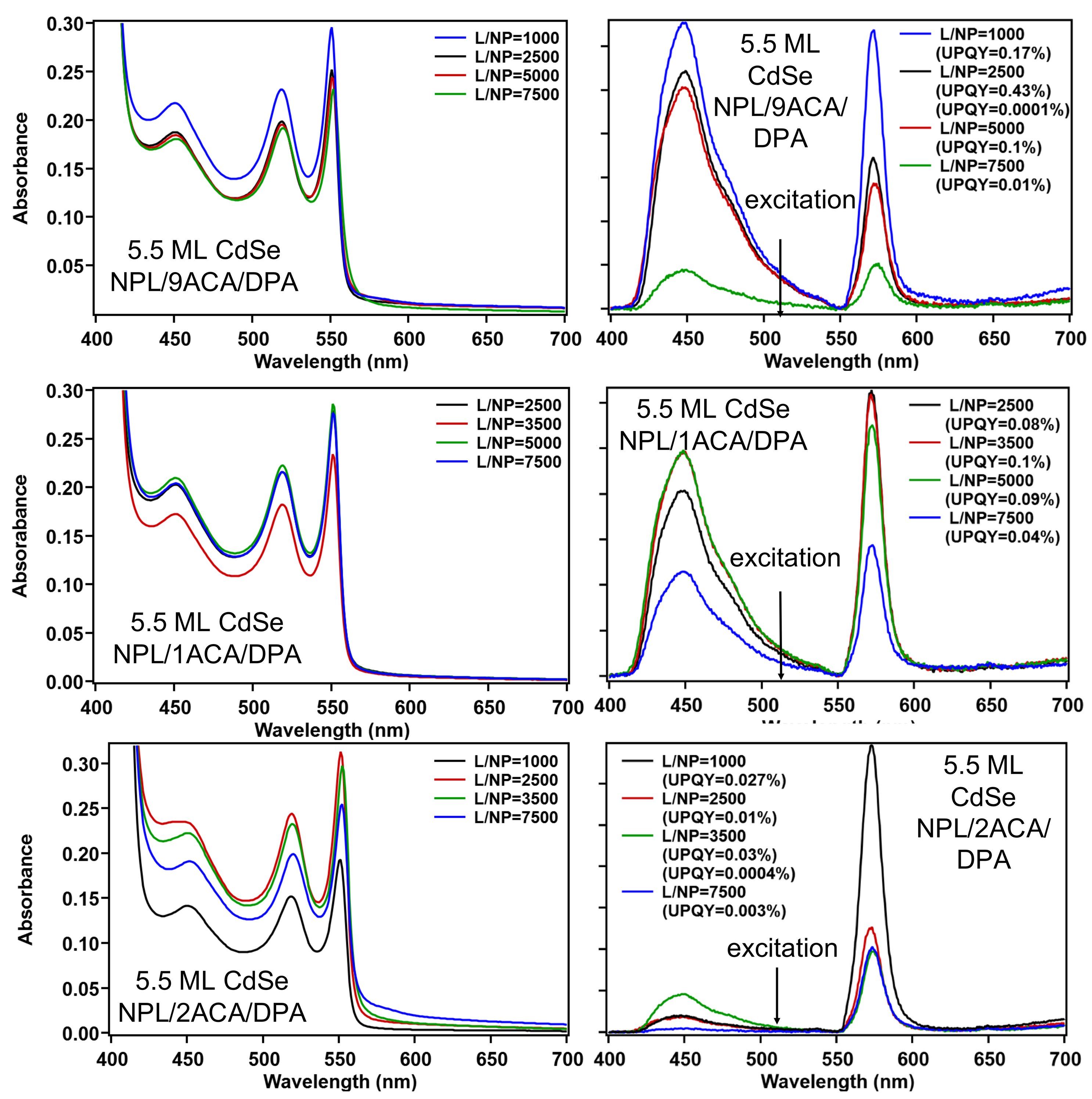
- Light absorber: nanoplatelets (NPL)
- Mediator: covalently bound to the NPL surface mediates energy transfer
- Emitter: acene molecules via triplet-triplet annihilation (TTA)

Nanoplatelets (NPL) as Sensitizers

- Quantum confinement in only one direction: erasing effects stemming from size polydispersity
- Enhanced exchange interactions: facilitate TET as the equilibrium tends to favor the dark state
- Large absorption cross sections

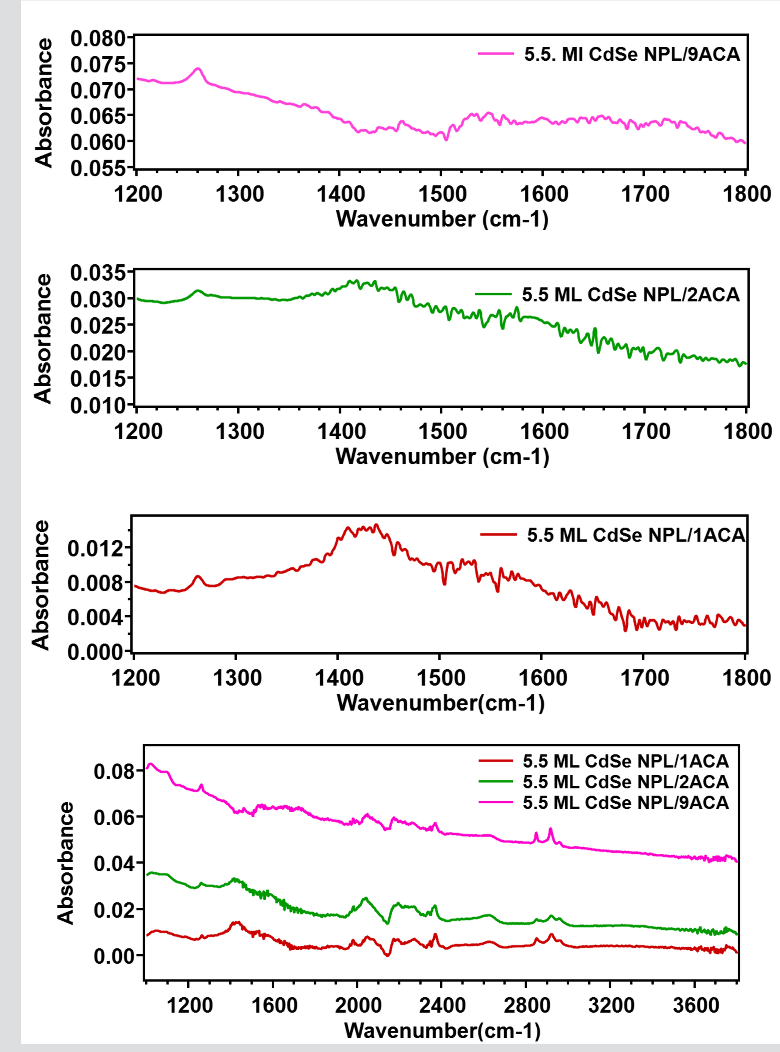


Varying Ligand to Nanoparticle Ratio



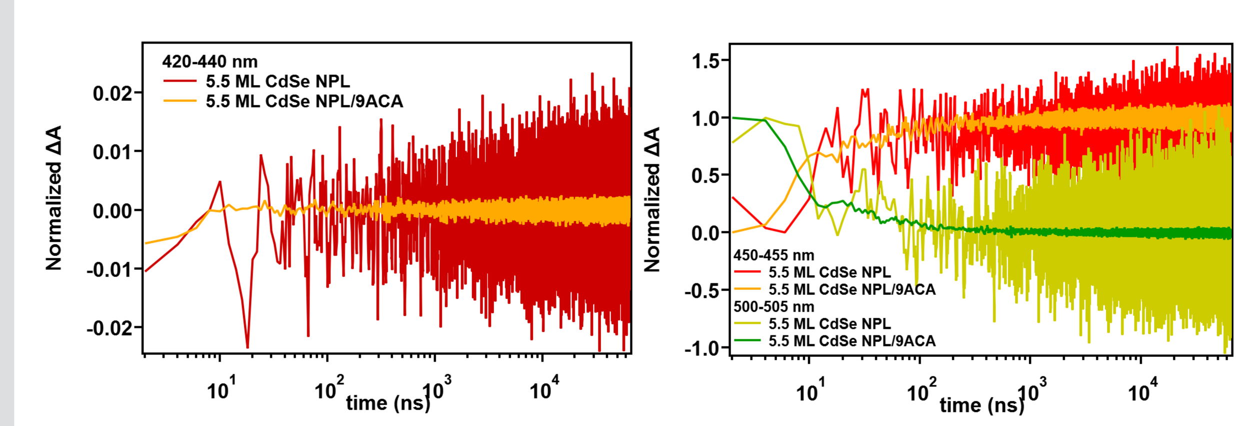
- The samples precipitated with high ligand loading for 2ACA, and with low ligand loading for 1ACA.
- We did not wash our samples with solvents to remove the excess ligands to prevent from crashing out.
- The highest upconversion quantum yield was 0.43% for 9ACA, 0.1% for 1ACA, and 0.03% for 2ACA with 3 mM diphenyl anthracene (DPA) in toluene at RT.

Attenuated Total Reflectance (ATR) Spectroscopy



- We washed the CdSe NPL/9ACA, CdSe NPL/1ACA, CdSe NPLACA with EtOH to remove excess ligands, and dry under the vacuum with LN2 to remove all the solvent.
- The sediment is re-dispersed in the small amount of hexane to perform ATR-IR.
- All samples show the peak at 1280 cm^{-1}.
- CdSe NPL/1ACA and CdSe NPL/2ACA show peaks at 1420 cm^{-1}, while CdSe NPL/9ACA doesn't.

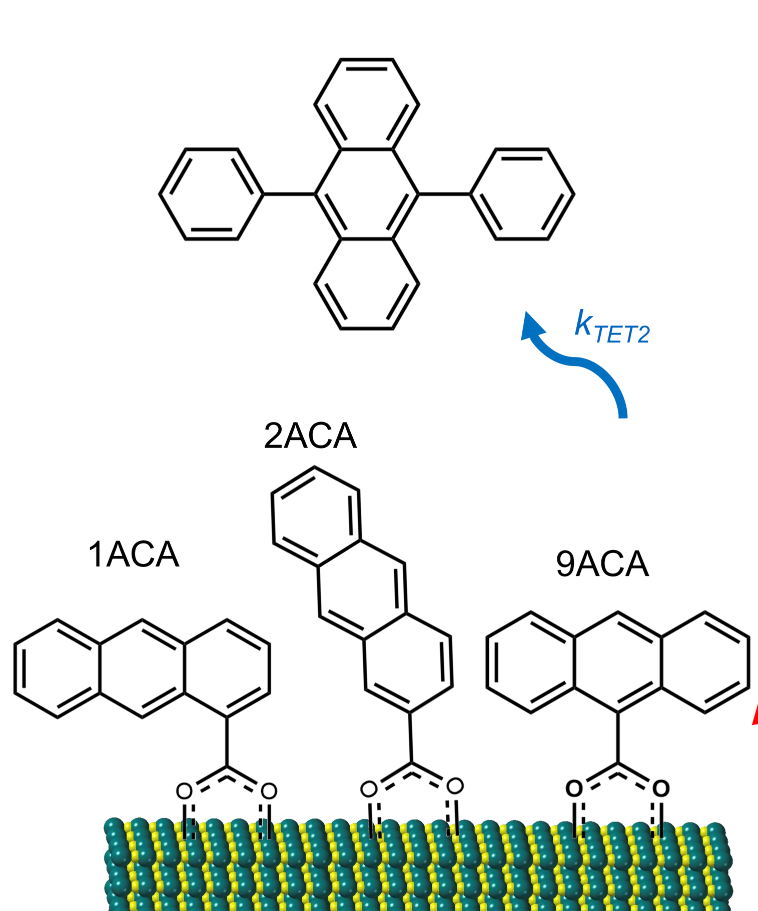
Transient Absorption: Kinetics



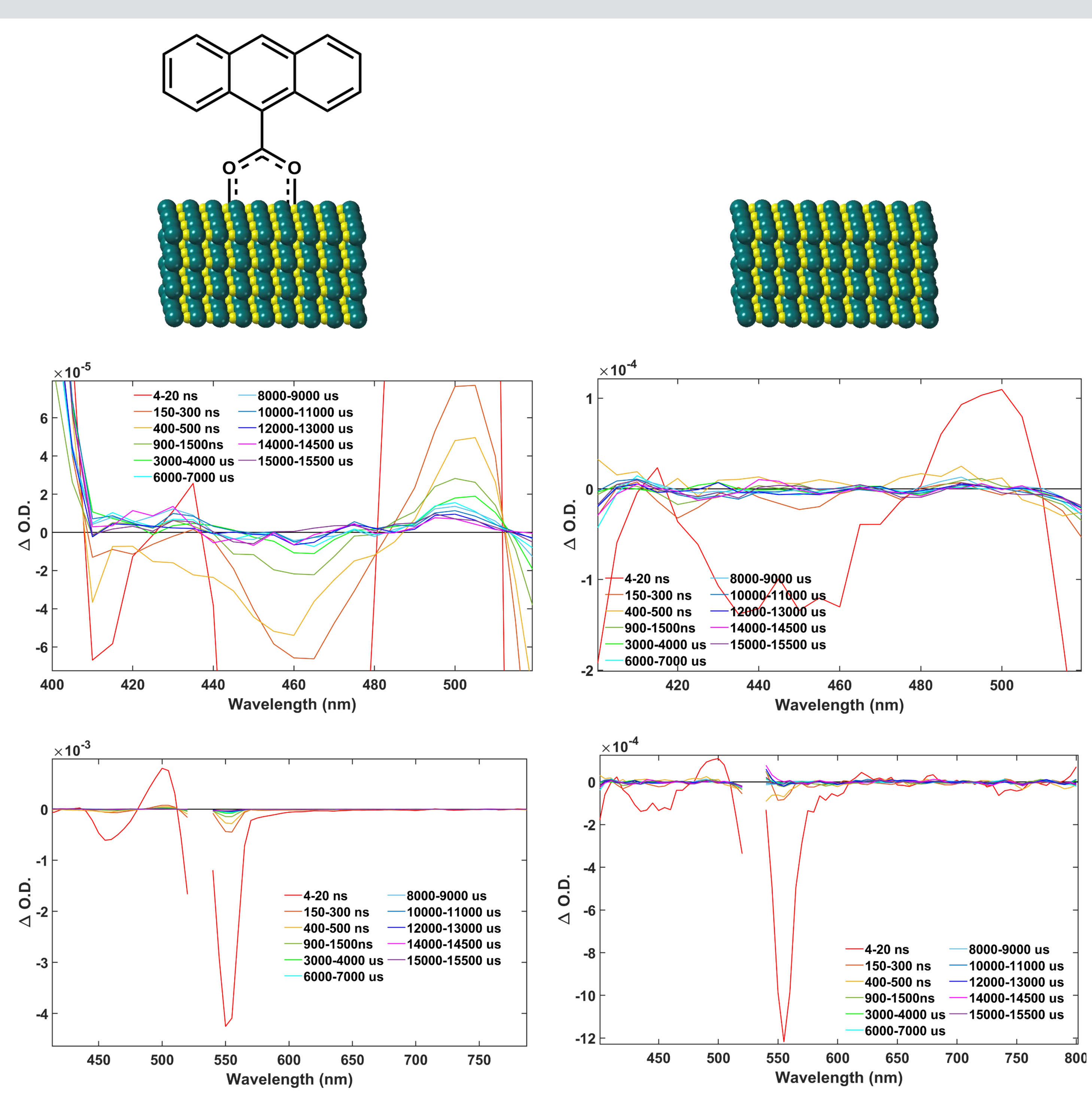
- After 9ACA is attached to 5.5 ML CdSe NPL, the kinetics at 450-455 nm and 500-505 nm are slightly quenched although our ns-TA IRF is 4.2 ns.
- There is no obvious 9ACA triplet ESA growth from 420-440 nm kinetics.
- This low triplet signal attributes to the low photon upconversion quantum yield (0.43%).

Comparing Electronic Coupling

- Comparing the electronic coupling of three isomers of anthracene-based transmitter ligands with CdSe nanoplatelet photosensitizers
- 1ACA, 2ACA, 9ACA mediator:1 Anthracenecarboxylic acid, 2 Anthracenecarboxylic acid , 9 Anthracenecarboxylic acid
- DPA emitter: 9, 10 diphenylanthracene



Nanosecond Transient Absorption (ns-TA) Spectroscopy



- There is no 9ACA excited state absorption (ESA) at triplet 430 nm at later times for 5.5ML CdSe NPL/9ACA sample.
- Both samples with and without 9ACA show ground state bleach (GSB) at 450 and 560 nm and ESA at 490 nm from CdSe NPL at early time.

Future Work

Enhancing Upconversion Quantum Yield

- Optimize Nanoplatelets Stacking: control of nanoplatelet stacking improves energy transfer and photoluminescence, enhancing quantum yield.
- Adjust Ligand-to-Nanoparticle Ratio: Optimizing this ratio enhances boost energy transfer efficiency and quantum yield.

References

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Acknowledgements

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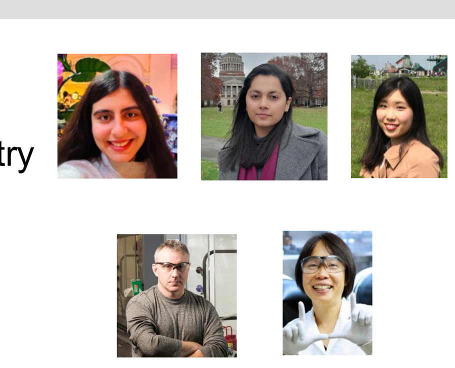

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The Tang group: Designer Nanostructures/ Ligands for Enhancing Energy Transfer